A Spectral Grouping and Attention-Driven Residual Dense Network for Hyperspectral Image Super-Resolution

Denghong Liu, Jie Li, *Member, IEEE*, and Qiangqiang Yuan, *Member, IEEE*

***Abstract*— Although unprecedented success has been achieved in convolutional neural network (CNN)-based super-resolution (SR) for natural images, hyperspectral image (HSI) SR without auxiliary high-resolution images remains a challenging task due to the high spectral dimensionality, where learning effective spatial and spectral representations is of great importance. In this article, we introduce a novel CNN-based HSI SR method, termed spectral grouping and attention-driven residual dense network (SGARDN) to facilitate the modeling of all spectral bands and focus on the exploration of spatial-spectral features. Considering the block characteristic of HSI, we employ group convolutions in and between groups composed of highly similar spectral bands at early stages to extract informative spatial features and avoid spectral disorder caused by normal convolution. To exploit spectral prior, a new spectral attention mechanism constructed by covariance statistics of features is designed to adaptively recalibrate features. We adapt the spectral attention for group convolutions to rescale grouping features with holistic spectral information. These two sequential operations called spectral grouping and integration module aim to extract effective shallow spatial-spectral features that are reused in the following layers. On the other hand, the residual dense block can better deal with spatial–spectral features by experimental comparison and hence is combined with the spectral attention to form a new basic building block for powerful feature expression and spectral correlation learning. The experimental results on synthesized and real-scenario HSIs demonstrate the feasibility and superiority of the proposed method over other state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), group convolution, hyperspectral image (HSI), spectral attention mechanism, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL sensors acquire images of the same scene in many contiguous bands over a certain electromagnetic spectrum. The resulting hyperspectral image (HSI) is a data cube with high spectral resolution consisting of a set of images recording the reflectance or radiance in different wavelengths [1]. With the availability of rich and detailed spectral information, the HSI has been extensively exploited in a variety of applications, such as classification [2], [3], target detection [4], geological exploration [5], and environmental monitoring [6]. However, mainly due to the limited amount of photons reaching each narrow spectral bands, HSIs are usually obtained under low spatial resolution incapable of capturing the details, which impedes their practical applications greatly. Therefore, many efforts are devoted to digital image processing technology, which is economical and effective without the hardware modification to recover spatial details from the observed HSI.

In general, the mainstream methods to enhance the spatial resolution of HSI are image fusion and single image super-resolution (SR). For the former, the higher spatial resolution image of the same scene, such as panchromatic image (PAN) and multispectral image (MSI), is required as a prior [7]–[10]. Since the auxiliary co-registered image can be unavailable in a real scenario, here we will focus on the progress of the latter that is more flexible and applicable without additional data. Single image SR is a signal postprocessing technology that aims to generate a high-resolution (HR) image from its low-resolution (LR) counterpart. Since it is an ill-posed inverse problem, conventional methods mainly focus on the utilization of effective spatial or spectral statistical distributions as priors, such as total variation, sparse regularization, low rank, and self-similarity to describe edges and textures [11]–[13]. However, these methods are usually time-consuming with sophisticated priors for good performance.

Currently, the convolutional neural network (CNN) has gained increasing research attention in HSI SR following the success of that in natural images (gray and RGB images). The pioneering work, SRCNN [14] has demonstrated the superiority of CNN over conventional methods to model a complex mapping between LR and HR image information. It was then further improved mainly by increasing network depth, which can widen the receptive field to exploit more contextual information and thus allow complicated modeling. Typical building blocks of a network that take advantage of skip connections, such as residual block (RB) [15], dense block (DB) [16], and residual DB (RDB) [17] are widely used to adapt the network to deeper layers and ensure good performance of SR.

Manuscript received August 8, 2020; revised October 13, 2020 and December 12, 2020; accepted January 3, 2021. Date of publication January 20, 2021; date of current version August 30, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 41922008, Grant 61971319, and Grant 62071341 and in part by the Hubei Science Foundation for Distinguished Young Scholars under Grant 2020CFA051. *(Corresponding author: Qiangqiang Yuan.)*

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Digital Object Identifier 10.1109/TGRS.2021.3049875

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As for CNN-based HSI SR, the most intuitive thought is to directly apply the CNNs for the SR of natural images to HSIs in a band-by-band manner [1], [18]. However, HSI SR is inherently more challenging than its natural image counterpart, mainly for the following reasons. On the one hand, the high spectral dimensionality of HSI contributes to reflect subtle spectral properties of measured objects but increases the difficulty of feature extraction. The bandwise manner with no concern for spectral correlation would result in spectral distortion [19]. Hence, recent researches pay more attention to spectral fidelity by imposing constraints, such as spectral difference learning [20], 3-D convolution [19], and spectral angle loss [21]. Simultaneous modeling of all spectral bands can yet hinder the representational power of CNNs. Fusing local spatial information of all spectral bands by normal convolution with the same weight can cause spectral disorder without discriminating differences of spectral bands [3], [22]. On the other hand, the spatial resolution of HSI is much lower than that of natural images. That is, the HSI may cover the area of a whole town, while an equally sized natural image depicts a single object with superior detail [18]. This can affect the quality of results due to insufficient detailed information. To fully explore the design of network architecture is a key factor affecting performance to meet this challenge.

With all the above considerations in mind, this article presents a novel end-to-end trainable SR network for HSI, termed spectral grouping and attention driven-residual dense network (SGARDN). Taking the 3-D HSI as input, the spectral grouping and integration module (SGIM) is designed to fully extract informative shallow spatial-spectral features from redundant spectral signatures at earlier stages for improving reconstruction ability. Specifically, the input LR HSI is firstly automatic subspace partitioned based on spectral correlation. Together with group convolutions, the intra- and inter-group feature extraction is achieved by exploring the correlations in and between these subspaces. Unlike previous methods achieve interactions across all spectral bands by normal convolution, the spectral attention mechanism is developed for grouping features to guide spectral integration by adaptively rescaling features with holistic spectral information. As for deep spatial–spectral features, several RDBs equipped with the spectral attention are stacked to exploit the correlations across spatial and spectral dimensions of HSI. We explore the residual dense network design for characterizing and combining abundant spatial-spectral features under different receptive fields. The building block RDB enables diversified and fine feature expression to handle challenge details by sharing and fusing the local features extracted from multiple preceding convolution layers. To allow the flexibility in dealing with different types of information, we further add the spectral attention before the local residual learning of RDB to selectively emphasize informative parts, which can exploit global spatial-spectral context to model spectral interdependencies and thus enable feature interactions to contribute differently to reconstruction. Considering the spectral correlation of HSI can be described by the correlation coefficient, the aforementioned spectral attention module is devised by exploiting the covariance statistic of features for making use of holistic spectral information at earlier stages and exploring spectral correlations of intermediate layers. As covariance is a kind of second-order statistics, it enables to learn more discriminative representations than first-order ones (e.g., maximum and average). Within this module, we apply depthwise convolution to the covariance statistic to respect its inherent structural information and adaptively find more suitable descriptors for each spectral band. Finally, features from various modules are aggregated to make full use of hierarchical features and improve the flow of information. The following deconvolution layers complete and speedup final reconstruction.

In summary, the main contributions of this article can be summarized as follows.

1) A novel CNN-based SR method for HSI is proposed. To exploit the characteristics across spatial and spectral dimensions of HSI, the RDB equipped with powerful spectral attention is constructed as the building block, which can perform advanced feature extraction and explore spectral correlations.
2) Considering the importance of low-level features for reconstruction, an SGIM is designed to extract informative shallow spatial–spectral features. It achieves not only the intra- and inter-group feature extraction but also the spectral integration with holistic spectral information.
3) For better mining spectral priors to facilitate the spatial enhancement and spectral consistency, a spectral attention module is devised to adaptively rescale features by using the covariance statistic that describes spectral correlation as spectral descriptors. We adapt it for group convolution and RDB to enable the network to make use of holistic spectral information at earlier stages and explore spectral correlations of intermediate layers.

The remainder of this article is organized as follows. In Section II, we introduce essential concepts related to our method in the context of existing approaches. Section III describes the newly proposed SGARDN method in detail. The analysis of network structure and experimental results are presented in Section IV. Finally, conclusions are given in Section V.

II. Related Work

A. CNN-Based HSI SR

With the rapid development of CNNs in natural images SR, early works are borrowed from them to HSIs in a band-by-band manner. Although they may be helpful in spatial resolution, they fail to consider spectral correlation, resulting in spectral distortion. Xie *et al.* [23] and Hu *et al.* [24] further integrated abundant spectral information of LR HSI through nonnegative matrix factorization (NMF). Li *et al.* [20] put forward to learn a spectral difference mapping between neighboring bands of LR HSI to HR HSI, together with certain spatial constraints that correct the spatial error to achieve better results. However, these methods need two stages to accomplish the reconstruction of HSI, which is inflexible and unstable. Mei *et al.* [19] constructed a three-dimensional full convolutional neural network (3-D FCNN), where 3-D convolution

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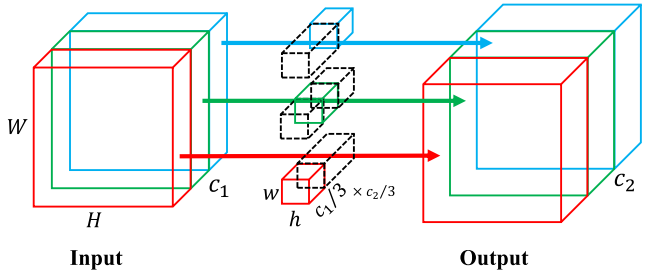


Fig. 1. c_1-channel input feature map of size $H \times W$ pixels trisected along channel dimension is convolved with three independent groups of $c_2/3$ filters of size $h \times w \times c_1/3$, respectively, to create a total c_2-channel output feature map.

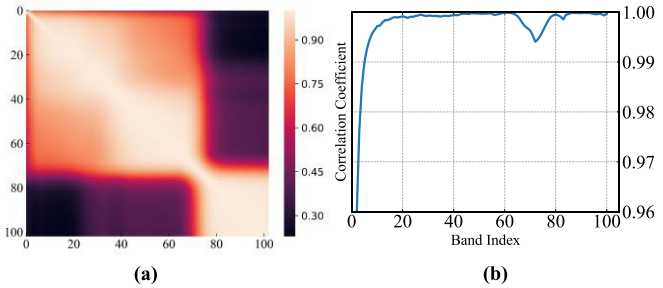


Fig. 2. Spectral correlation analysis for Pavia Centre. (a) Visualization of correlation matrix. (b) Correlation coefficient between current band i (the value on the x-axis) and $i + 1$. The larger the correlation coefficient, the stronger the correlation.

operation is used to explore the spatial context and spectral correlation jointly. Although it is effective in alleviating spectral distortion, its computational complexity is very high. Li *et al.* [21] developed a grouped recursive module of 2-D convolution, enabling the network to be compact. In addition, they conjoin the spatial error with the spectral angle mapper (SAM) loss to keep the spectral signature. To prevent the spectral disorder caused by normal 2-D convolution which quickly fuses the features of all channels, Zheng *et al.* [22] proposed the separable-spectral convolution to independently extract features of each spectral band. However, it would produce so many redundant features due to hundreds of spectral bands that it is hard to identify the most representative parts.

B. Group Convolution

The concept of group convolution first appeared in AlexNet [25] for highly optimized implementation on multiple GPUs. As depicted in Fig. 1, this scheme subdivides input feature maps into several groups, within which independent convolution is performed. The sparse connection structure enforced by group convolution allows a significant reduction in computational cost and number of parameters without compromising or even improving the accuracy [26]. Later, the depthwise separable convolution is widely used for the design of lightweight models [27]–[29]. It consists of a depthwise convolution and a 1×1 convolution called pointwise convolution for combining. The depthwise convolution is performed over each input channel independently, that is, the number of groups equals that of input channels.

Aside from the above virtues, we demonstrate that the HSI can benefit from the employment of group convolution given its inherent characteristics. In this light, we calculate the pairwise correlation matrix between all bands to display their relationships with each other. As depicted in Fig. 2(a), we can observe the so-called block characteristic of HSI. That is, the adjacent bands contain redundant information. Therefore, it is plausible that group convolution can deal with the high dimensionality of HSI to extract informative features from the band subspaces divided according to the correlation matrix.

C. Attention Mechanism

The attention mechanism, derived from the fact that the human vision selectively focuses on salient parts instead of processing a whole scene at once, is a prevalent practice of CNNs nowadays that stands out in various computer vision tasks through adaptive feature refinement. Hu *et al.* [30] introduced the squeeze-and-excitation (SE) block to explicitly model the inter-channel dependencies for image classification. Park *et al.* [31] and Woo *et al.* [32] computed the complementary attention along the channel and spatial axes sequentially to further boost the representation power of CNNs. Generally, the attention mechanism can be divided into two steps, squeeze and excitation, which first extracts statistics from global information mainly with first-order pooling methods, and then captures interrelationships by convolutions followed by nonmutually exclusive activation functions to recalibrate the input feature maps [30].

To pay more attention to informative features such as high-frequency information conducive to HR reconstruction, some SR methods have applied the attention mechanism that was mainly borrowed from other vision problems [33], [34]. Kim *et al.* [35] devised a new attention method that is optimized for SR with consideration of the different nature between SR and other problems. Recently, there have been several attempts to explore statistics higher than first-order for enhancing nonlinear modeling capability [36], [37]. Dai *et al.* [38] proposed a second-order channel attention (SOCA) module to learn more discriminative representations by considering second-order feature statistics, which has been shown to be effective for SR.

Due to the competitive advantages in the representational capacity, some efforts have been devoted to adapt the attention mechanism to HSI SR [39], [40]. These works simply explore first-order statistics (e.g., global average pooling) as image representations. However, since the spectral correlation of HSI can be described by the correlation coefficient or normalized covariance, second-order statistics such as the covariance matrix are a promising way to explore spectral characteristics. To this end, we propose a second-order attention mechanism that enables feature interactions through spectral correlation.

III. Methodology

In this section, the proposed SGARDN for HSI SR is presented in detail. It aims to learn an end-to-end mapping function $H(\cdot)$ with parameters θ by fully exploring spatial-spectral information between the LR image I_{LR} and the HR image I_{HR} with a size of $h \times w \times c$ and $rh \times rw \times c$,

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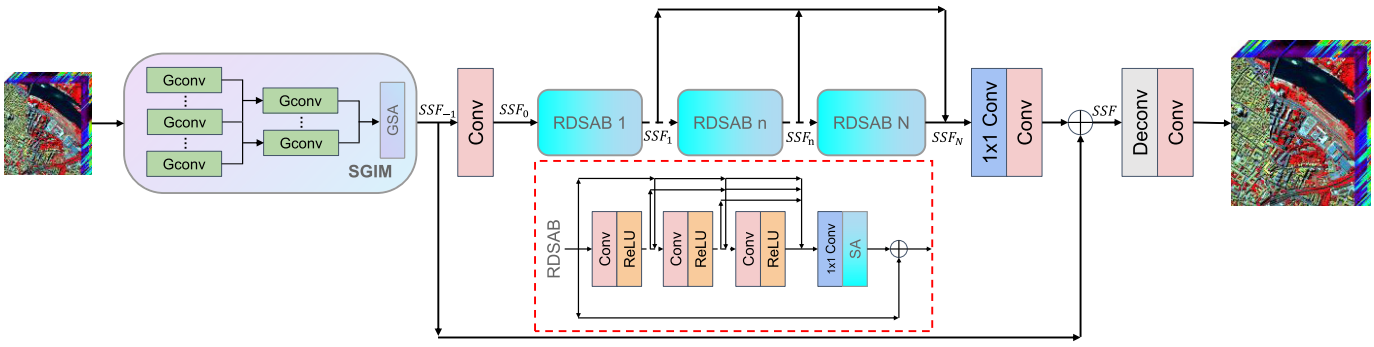


Fig. 3. Overview of our proposed SGARDN. Key components of the network are visually displayed: 1) the SGIM extracts informative shallow spatial-spectral features, composed by group convolutions (GConv) and the spectral attention for grouping features (GSA) and 2) the RDBs equipped with the spectral attention (RDSAB) are stacked to produce deep hierarchical spatial-spectral features for accurate reconstruction.

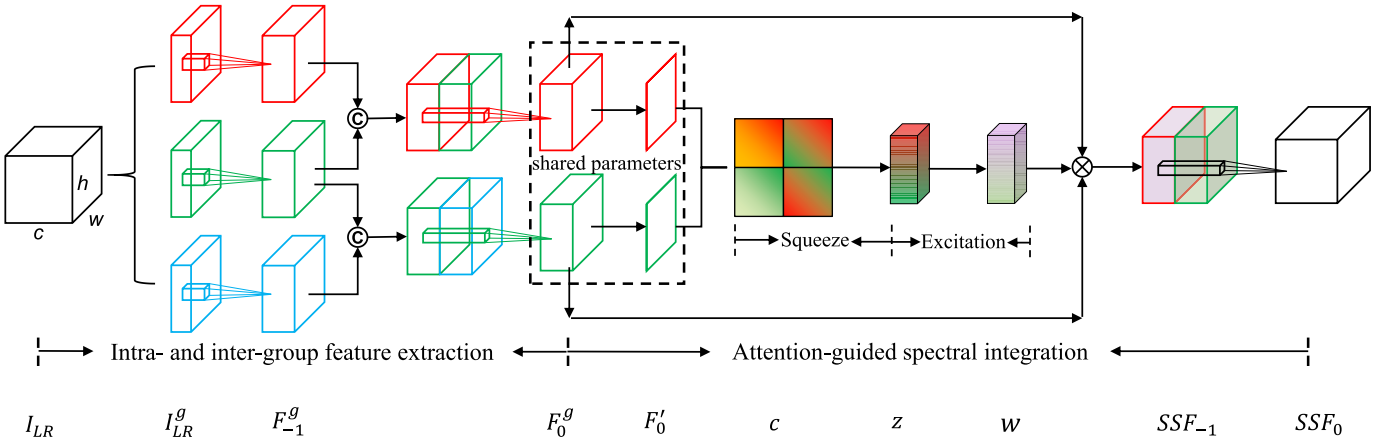


Fig. 4. Structure of the SGIM.

respectively, where h, w and c represent the height, width, and (spectral) channel number and r denotes the scale factor. The super-resolved result I_{SR} similar to I_{HR} can be reconstructed through

$$I_{\mathrm{SR}} = H(I_{\mathrm{LR}}; \theta). \tag{1}$$

A. Network Structure

As shown in Fig. 3, the architecture of SGARDN mainly consists of three parts: SGIM, residual dense spectral attention block (RDSAB), and final reconstruction (REC) subnet. The first part aims to extract informative spatial–spectral features SSF_0. The input LR image I_{LR} is automatic subspace partitioned based on spectral correlation and group convolutions are employed in and between subspaces for spatial feature extraction. These grouping features are then recalibrated with holistic spectral information through the spectral attention mechanism followed by a convolutional layer. The above process can be formulated as follows:

$$\mathrm{SSF}_0 = H_{\mathrm{SGIM}}(I_{\mathrm{LR}}) \tag{2}$$

where H_{SGIM} represents a composite operation of group convolutions, spectral attention, and convolution, the diagram of which is presented in Fig. 4.

The next part contains N RDSABs operating on SSF_0 to produce deep spatial–spectral features. Specifically, the output SSF_n of the nth RDSAB can be obtained by

$$\begin{aligned}\mathrm{SSF}_n &= H_{\mathrm{RDSAB},n}(\mathrm{SSF}_{n-1}) \\ &= H_{\mathrm{RDSAB},n}\left(H_{\mathrm{RDSAB},n-1}\left(\cdots\left(H_{\mathrm{RDSAB},1}(\mathrm{SSF}_0)\right)\cdots\right)\right)\end{aligned} \tag{3}$$

where $H_{\mathrm{RDSAB},n}$ stands for the operations of nth RDSAB, which can be a composite function formed by convolutions, rectified linear units (ReLUs), and spectral attention. More details will be given later.

Since there is complementary information among different levels of feature maps, the last part first concatenates hierarchical spatial–spectral features generated in the preceding modules, expressed as $[\mathrm{SSF}_1, \ldots, \mathrm{SSF}_N]$, followed by a 1×1 convolution and a 3×3 convolution, which can adaptively control the output information while reducing the number of feature maps and further conduct global residual learning, to obtain the fused spatial-spectral features SSF for final reconstruction as follows:

$$I_{\mathrm{SR}} = H_{\mathrm{REC}}(\mathrm{SSF}) \tag{4}$$

where H_{REC} is made up of deconvolution layers and a convolutional layer, which can vary with the scale factor. For instance, $\times 4$ network has two successive deconvolution layers.

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Recent works commonly employ deconvolution [41] or subpixel convolution [42] as a postupscaling strategy to learn a set of upscaling filters instead of a pre-upscaling strategy with fixed interpolation, which achieves higher performance while decreasing computational cost quadratically by performing SR in LR space. The subpixel convolution rearranges the elements by mapping pixels along the channel dimension to the spatial domain to increase the size at the specified scale factor, that is, it needs to generate r^2c feature maps. This would be enormous due to the high spectral dimensionality of HSI. Therefore, we utilize deconvolution for enlarging, which is considered as the inverse operation of convolution.

B. Spectral Grouping and Integration Module

Low-level features at early stages are significant to the network as they are reused by subsequent layers. That is, effective low-level features can facilitate reconstruction ability. It is for this reason that we design the SGIM for shallow spatial-spectral feature extraction. This module is outlined in Fig. 4, consisting of two parts: intra- and inter-group feature extraction and attention-guided spectral integration.

1) Intra and Inter-Group Feature Extraction: Since HSI with high spectral dimensionality contains a lot of redundant information, it is sensible to partition it into appropriate subspaces for better extracting features. The common practice achieves this division in accordance with the spectral correlation property expressed by correlation coefficients as follows:

$$R_{i,j} = \frac{E\left\{\left(I_{\mathrm{LR}}^{i} - \overline{I}_{\mathrm{LR}}^{i}\right)\left(I_{\mathrm{LR}}^{j} - \overline{I}_{\mathrm{LR}}^{j}\right)\right\}}{\sqrt{E\left\{\left(I_{\mathrm{LR}}^{i} - \overline{I}_{\mathrm{LR}}^{i}\right)^2\left(I_{\mathrm{LR}}^{j} - \overline{I}_{\mathrm{LR}}^{j}\right)^2\right\}}} \tag{5}$$

where i, j denote the index of bands, and superscript, E represent the mean values and mathematical expectation, respectively. According to (5), the correlation matrix $R \in \mathbb{R}^{c\times c}$ of all spectral bands can be obtained, from the visualization of which [Fig. 2(a)] the block characteristic is obvious. We then follow [43] to implement automatic subspace partition based on the continuity and transmissibility of the local correlation of neighboring bands. Specifically, local minimum values of the $(c-1)$ neighboring transmissible correlation coefficients are used as thresholds [e.g., there are two local minima in Fig. 2(b)]. Thus, the I_{LR} is subdivided into G subspaces written as

$$I_{\mathrm{LR}} = \left[I_{\mathrm{LR}}^{1}, \ldots, I_{\mathrm{LR}}^{g}, \ldots, I_{\mathrm{LR}}^{G}\right] \tag{6}$$

where brackets $[\cdots]$ refer to the operation of concatenation throughout this article.

Each subspace generally has a different dimension, but similar properties internally. In order to fully exploit intra-subspace spatial features, a group convolution GC_1 is employed, in which each subspace corresponds to independent convolution

$$F_{-1} = \mathrm{GC}_1(I_{\mathrm{LR}}) \tag{7}$$

where F_{-1} in the form of $[F_{-1}^{1}, \ldots, F_{-1}^{g}, \ldots, F_{-1}^{G}]$ is grouping features.

From the perspective of the group, neighboring groups still have close connections to some extent, so a group convolution GC_2 further works on the concatenation of two sequential grouping features. It enables inter-group interactions to further extract informative spatial features. Meanwhile, it reduces the redundancy of feature maps and the computational cost of the succeeding operations

$$F_0^g = \mathrm{GC}_2^g\left(\left[F_{-1}^{g-1}, F_{-1}^{g}\right]\right) \tag{8}$$

where the gth grouping features F_0^g is generated by gth independent convolution of GC_2 on the concatenated features of two neighboring groups. Note that there are totally $G-1$ groups in feature F_0.

Besides, normal convolution sums over all spectral bands of input, and hence spectral dependencies are implicitly embedded in the feature maps, which will be to the disadvantage of modeling the HSI, since as shown in Fig. 2(a), there is a clear distinction for bands further apart. Therefore, normal convolution will cause spectral disorder by tackling all spectral bands at once, which can be avoided by group convolution.

2) Attention-Guided Spectral Integration: For the sake of spectral integrality, it is necessary to access the overall information of all spectral bands. As mentioned above, normal convolution can fuse the features of each spectral band, but the spectral disorder is inevitable. Therefore, the attention mechanism is employed to recalibrate features with holistic image information. Following generic pipelines, there are two steps, namely squeeze and excitation, to generate the spectral descriptor, and capture bandwise dependencies, respectively. The right part of Fig. 4 shows the diagram of our attention module for grouping features.

Since the features within groups are obtained based on local properties of HSI, we first employ convolution for global adaptive pooling AptPool to produce representative feature maps of each group by embedding local spectral information. It achieves dimension reduction at the same time to significantly decrease the computational complexity of subsequent operations. All groups share the same parameters to enable the aggregated features to lie in the same semantic embedding space

$$F_0' = \mathrm{AptPool}(F_0). \tag{9}$$

Note that the global adaptive pooling along the spectral dimension operates on each group independently to obtain representative features $F_0' \in \mathbb{R}^{h\times w\times(G-1)}$.

We then explore the covariance matrix, which reveals the interrelationships among spectral bands and has strong modeling capacity with more discriminative representations

$$c = \mathrm{COV}\left(F_0'\right) \tag{10}$$

where COV computes the pairwise spectral covariance to form the covariance matrix $c \in \mathbb{R}^{(G-1)\times(G-1)}$, each row of which indicates the statistical dependence of a specific band with all spectral bands. Similar to [37], the covariance matrix is then reshaped to a 3-D tensor $c \in \mathbb{R}^{1\times(G-1)\times(G-1)}$ that means $1\times(G-1)$ feature map with $(G-1)$ channels for simplification of succeeding operations, followed by row-wise normalization

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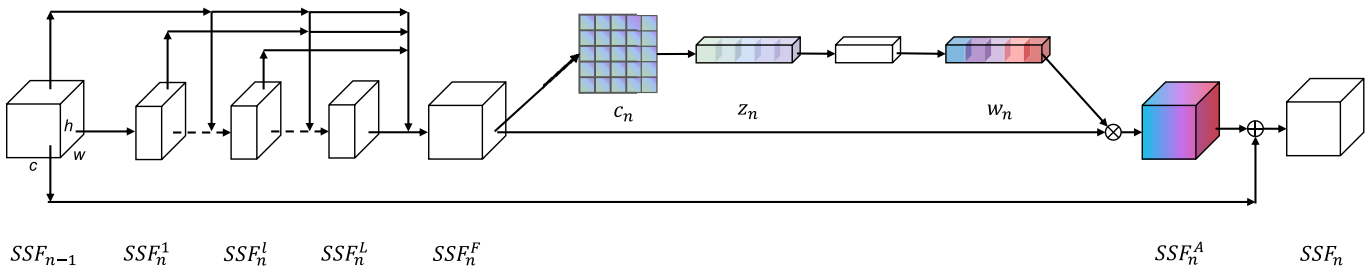


Fig. 5. RDSAB architecture.

to keep the order of data. A depthwise convolution $\text{Conv}_{\text{depth}}$ operate on the covariance matrix to generate spectral descriptors $z \in \mathbb{R}^{1\times1\times(G-1)}$. Specifically, it applies a single filter with the size of $1 \times (G-1)$ to each band in view of the inherent structural information, which can adaptively find more suitable descriptors for each spectral band, unlike the fixed average pooling

$$z = \text{Conv}_{\text{depth}}(c). \tag{11}$$

To fully capture spectral dependencies from the information aggregated in z, a convolution W with nonmutually-exclusive activation function sigmoid σ is employed to project the descriptors onto weights $w \in \mathbb{R}^{1\times1\times(G-1)}$ of each group

$$w = \sigma(W(z)). \tag{12}$$

Finally, channel-wise multiplication between the weights w and the corresponding grouping features F_0 is performed to obtain SSF_{-1}, which can be seen as shallow spatial–spectral features due to the contact with holistic spectral information. SSF_{-1} is used for global residual learning later to pass abundant low-level features, which has been demonstrated to be effective in [44]. We then generate shallow feature SSF_0 for further deep feature extraction by 1×1 convolution, which enables feature interactions and reduces the number of feature maps.

C. Residual Dense Spectral Attention Block

As shown in Fig. 3 (the contents in the red dotted box) and Fig. 5, the proposed RDSAB integrates the spectral attention mechanism into RDB that can perform advanced feature extraction, benefited from skip connections. We also experimentally show that RDB can deal with the spatial–spectral features more effectively compared with other commonly used blocks in Section IV-B. Each convolutional layer in current RDSAB has access to not only all the subsequent layers but also the output of preceding RDSAB, resulting in the contiguous memory mechanism [17]. Since the size of the receptive field is proportional to the depth, a set of RDSABs are stacked for learning hierarchical spatial–spectral features under different receptive fields. The output SSF_n^l of the lth convolutional layer of n-th RDSAB can be formulated as follows:

$$\text{SSF}_n^l = H_{\text{RDSAB},n}^l\left(\left[\text{SSF}_{n-1}, \text{SSF}_n^1, \ldots, \text{SSF}_n^{l-1}\right]\right) \tag{13}$$

where $H_{\text{RDSAB},n}^l$ denotes a convolution operation followed by a ReLU activation function. Denote K as the growth rate, namely the number of feature maps produced by each convolution layer. And the output of nth RDSAB has K_0 feature maps. The dense skip connections in RDSAB enable the long-term dependencies and enhance feature propagation, while the concatenation of feature maps combines features of different levels, thereby increases the variation of the input in subsequent layers to boost reconstruction performance [45].

As concatenating all feature maps of preceding layers will yield many inputs for the subsequent layer, it is essential to reduce the number of feature maps. The local fused output SSF_n^F of nth RDSAB can be generated through

$$\text{SSF}_n^F = H_{\text{RDSAB},n}^{FF}\left(\left[\text{SSF}_{n-1}, \text{SSF}_n^1, \ldots, \text{SSF}_n^l, \ldots, \text{SSF}_n^L\right]\right) \tag{14}$$

where $H_{\text{RDSAB},n}^{FF}$ represents the feature fusion with 1×1 convolution. It is a common operation that can improve the model compactness while adaptively preserve more useful spatial–spectral features.

Here, we mainly have three concerns. First, the convolutional kernel treats all types of information equally, which can restrict the reconstruction ability because it is hard to distinct valuable high-frequency spatial details among abundant low-frequency information [35]. Second, the convolutional kernel only expresses connectivity patterns within local receptive fields, lacking enough contextual information. Third, since spectral information is of significance to HSI, the full exploration of spectral correlation cannot be ignored. Therefore, we integrate the spectral attention into RDB to assist with the identification of representative parts through holistic information. Covariance statistics are considered as spectral descriptors for spectral correlation learning. It is essentially the same spectral attention that was mentioned earlier for grouping features. Specifically, we compute pairwise spectral correlations to obtain the covariance matrix $c_n \in \mathbb{R}^{K_0\times K_0}$ and then employ a depthwise convolution on the normalized covariance matrix to generate a spectral descriptor $z_n \in \mathbb{R}^{1\times1\times K_0}$ adaptively in the squeeze process. We apply the gating mechanism to generate weight $w_n \in \mathbb{R}^{1\times1\times K_0}$ by fully capture bandwise dependencies in the excitation step. The gating mechanism is a bottleneck with two fully connected layers around the nonlinearity

$$w_n = \sigma\, W_U(\delta W_D(z_n)) \tag{15}$$

where δ and σ refer to the ReLU and sigmoid function, respectively. W_D is a dimensionality-reduction layer with a

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reduction ratio 16, while W_U is a dimensionality-increasing layer. Then, we use these weights to rescale the local fused output SSF_n^F by channel-wise multiplication to obtain SSF_n^A.

Finally, local residual learning further improves the information flow and network representation ability. The final output SSF_n of nth RDSAB can be obtained through

$$\mathrm{SSF}_n = \mathrm{SSF}_{n-1} + \mathrm{SSF}_n^A. \tag{16}$$

D. Loss Function

Given a training set $\{I_{\mathrm{LR}}^i, I_{\mathrm{HR}}^i\}_i^N$, where superscript i denotes the ith of total N training images, we minimize the following mean absolute error (MAE):

$$L_1 = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\mathrm{chw}}\left\|I_{\mathrm{SR}}^i - I_{\mathrm{HR}}^i\right\|_1. \tag{17}$$

It is noticeable that although the default mean squared error (MSE) loss function is in favor of peak signal-to-noise ratio (PSNR), it suffers from several limitations [46], such as the problems of convergence and over-smoothness. So, we adopt the MAE loss for training instead of MSE loss.

To ensure spectral consistency, there is a popular way to minimize the spectral angle [21] between the reconstructed and ground-truth spectra simultaneously as follows:

$$L_{\mathrm{spectral}} = \frac{1}{\mathrm{NN}_S}\sum_{i=1}^{N}\sum_{j=1}^{N_S}\arccos\left(\frac{I_{\mathrm{SR}}^{i,j}\cdot I_{\mathrm{HR}}^{i,j}}{\left\|I_{\mathrm{SR}}^{i,j}\right\|_2\left\|I_{\mathrm{HR}}^{i,j}\right\|_2}\right) \tag{18}$$

where superscript j denotes the jth of total N_S spectral vectors of each image.

We train our networks with L_1, and jointly with L_1 and L_{spectral}, respectively, where there exists a balance factor λ. The evaluation of this comparison is provided in Section IV-B.

IV. Experiments

In this section, we provide an in-depth experimental analysis of network structure and the comparison with other state-of-the-art methods on both synthesized and real-scenario data sets to verify the performance of the proposed method.

A. General Information

1) Data Sets: The performance of the proposed method is evaluated on both synthesized and real-scenario HSIs. For the synthesized HSIs, the CAVE data set [47] comprising 32 scenes of a wide variety of real-world materials and objects is employed. Each image has 512×512 pixels and 31 spectral bands with wavelengths ranging from 400 nm to 700 nm under the controlled illumination. We randomly choose four images for evaluation and the rest are used for training (10% of the training patch pairs are selected to form the validation set).

The real-scenario HSIs come from the Pavia Centre and Chikusei [48] remote sensing data. The former was acquired from a flight campaign over Pavia, northern Italy by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. It contains 1096×715 valid pixels with a geometric resolution of 1.3 m after discarding the samples with no information and 102 bands after removing water absorption bands. We select an original region with rich details of size 150×150, similar to the setting in [19], to validate the performance.

The latter was taken by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan. It contains 2517×2335 pixels with 128 bands in the spectral range from 363 to 1018 nm and the ground sampling distance was 2.5 m. By following the experimental setting in [48], we extract four nonoverlap regions with $512\times512\times128$ pixels to form the test data and the remaining regions are for training.

The above three data sets are considered as ground-truth of high spatial resolution HSIs, while the LR HSIs are simulated by bicubic downsampling (the opencv-python function resize) the original HSIs with different scaling factors.

2) Evaluation Criteria: Three prevailing evaluation criteria are utilized to comprehensively verify the proposed method on both spatial and spectral sides. To evaluate the spatial reconstruction quality of each spectral band at image levels, the PSNR and the structural similarity (SSIM) index [49] are adopted. The former describes the similarity based on MSE, while the latter measures structural consistency. We use their mean values over all spectral bands as spatial quality indicators of the whole HSI. Concerning the spectral reconstruction quality of each spectrum at pixel-levels, the SAM [18] is exploited. It evaluates the spectral fidelity by calculating the angle between spectral vectors of the reconstructed HSI and that of the ground-truth. We also use the mean value of SAM over all pixels as the spectral quality indicator of the whole HSI.

Given a ground-truth HSI $X\in\mathbb{R}^{B\times P}$ containing P pixels with B spectral bands and a reconstructed one $\widehat{X}$, these evaluation criteria can be defined as follows:

$$\mathrm{PSNR}\left(x_i,\hat{x_i}\right) = 10\log_{10}\left(\frac{\max(x_i)^2}{||x_i-\hat{x_i}||_2^2/P}\right) \tag{19}$$

$$\mathrm{SSIM}\left(x_i,\hat{x_i}\right) = \frac{\left(2\mu_{x_i}\mu_{\hat{x_i}}+c_1\right)\left(2\sigma_{x_i\hat{x_i}}+c_2\right)}{\left(\mu_{x_i}^2+\mu_{\hat{x_i}}^2+c_1\right)\left(\sigma_{x_i}^2+\sigma_{\hat{x_i}}^2+c_2\right)} \tag{20}$$

$$\mathrm{SAM}\left(x_j,\hat{x_j}\right) = \arccos\left(\frac{<x_j^T\hat{x_j}>}{||x_j||_2||\hat{x_j}||_2}\right) \tag{21}$$

where $x_i\in\mathbb{R}^{P\times1}$ and $x_j\in\mathbb{R}^{B\times1}$ are the ith band image and the spectral vector of jth pixel, respectively. max is for maximum operation, μ denotes the mean value, and σ denotes the variance or covariance. c_1 and c_2 are two constants to avoid instability when the denominator is close to 0. $<\cdot>$ represents the dot product of two vectors, and $||\cdot||_2$ represents the l_2 norm operation.

Generally, the larger the values of PSNR and SSIM, the better the spatial quality, whereas, the smaller the value of SAM, the less the spectral distortion.

3) Competing Methods: Seven strategies are selected as the baseline for comparison: Bicubic, SRCNN [14], SRDenseNet [16], RDN [17], 3DFCNN [19], GDRRN [21], and SSPSR [48]. Among these, Bicubic is a classic

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interpolation method; SRCNN, SRDenseNet, and RDN are the state-of-the-art CNN-based SR methods for natural images, and even more specifically, these methods are representative of different levels of depth (from several to more than a hundred layers) and network structure (plain structures or skip connections); 3DFCNN, GDRRN, and SSPSR are recent successful CNN-based SR methods designed for HSI. To make the CNNs of natural images SR suitable for HSI, a band-by-band manner is employed. That is, we treat the HSI as a series of independent gray images. All the competing methods are implemented via the publicly released codes. To be fair, we follow the network settings in corresponding papers and train them with the same training set until the loss converges. Specifically, SRCNN has three convolutional layers, and the kernel size is $9 \times 9 \times 64$, $1 \times 1 \times 32$, and $5 \times 5 \times 1$, respectively. SRDensenet uses eight DenseNet blocks with a growth rate of 16, resulting in 64 convolution layers. RDN has 16 RDBs, eight convolutional layers per RDB, and 64 filters per convolutional layer. 3DFCNN consists of four convolutional layers, and the kernel size is $9 \times 9 \times 7 \times 64$, $1 \times 1 \times 1 \times 32$, $1 \times 1 \times 1 \times 9$, and $5 \times 5 \times 3 \times 64$, respectively. Note that since 3DFCNN would change the number of spectral bands, we expand the original bands as the input by replicating the first and last few bands. GDRRN has nine recursive blocks, within which the kernel size of each layer is $3 \times 3 \times 128$. Note that the original convolutional layers in recursive blocks are replaced with group convolution (the number of groups is 2) to reduce more parameters. The branch and global network of SSPSR have three spatial-spectral blocks, respectively, and the kernel size of each layer is $3 \times 3 \times 256$ except the spectral residual modules within the block. We display the comparison of network parameters in Fig. 16 for an intuitive understanding.

4) Implementation Details: In our proposed network, the kernel size of convolutional layers is 3×3 unless specified, and the zero-padding is used to keep all feature maps in the same size. The output of each RDSAB is 128 feature maps. The basic parameters will be investigated experimentally later. The weights are initialized by the method proposed in [50]. The network is optimized using Adam optimizer [51] with the default setting. In each training batch, we randomly extract 32 LR HSI patches with the size of $32 \times 32 \times c$ cropped from the training images as inputs in consideration of the scale of the network. We randomly augment the patches by flipping horizontally or vertically and rotating 90° [17]. The initial learning rate is set at 1×10^{-4} and halved every 100 epochs, whereas fixed at 5×10^{-5} on the real-scenario HSIs. We implement the proposed network with the TensorFlow framework on an NVIDIA RTX 2080Ti GPU.

B. Network Structure Analysis

1) Study of Basic Building Block: Note that RDB here is without the continuous memory mechanism for a fair comparison, and the only difference between DB and RDB is the local residual learning. We construct five architecturally identical networks with roughly the same number of parameters but stacked by different blocks, which take the whole 3-D HSI as input. From the first three rows of Table I, we can verify the capability of basic blocks commonly used in CNNs. The RB performs worse than the others in both spatial and spectral criteria, which demonstrates that dense connections are more appropriate for dealing with the spatial-spectral features. When adding the local residual learning to DB (i.e., RDB), the performance is further improved by a large margin. This is consistent with our analysis in Section III-C. The last two rows of Table I show the effect of the attention mechanism. RDB_SE and RDB_SA denote the RDB equipped with squeeze and excitation block [30] and our proposed spectral attention, respectively. We find that the attention mechanism is conducive to the reconstruction and the proposed spectral attention is superior to the classic squeeze and excitation block for HSI by taking spectral correlation into consideration.

TABLE I
COMPARISON OF DIFFERENT BASIC BUILDING BLOCKS FOR SCALE FACTOR ×2 ON THE PAVIA CENTRE DATA SET

Basic block	PSNR	SSIM	SAM
RB	33.32	0.9945	4.2045
DB	33.36	0.9946	4.1065
RDB	33.50	0.9947	4.0687
RDB_SE	33.48	0.9947	4.0526
RDB_SA	**33.53**	**0.9948**	**3.9969**

2) Parameter Analysis: In this section, we give sound reasoning on the selection of parameters through various experiments. The basic parameters involved in the proposed network include the number of RDSAB (denote as N for short), the number of convolutional layers per RDSAB (denote as L for short), the growth rate of RDSAB (denote as K for short), and the number of filters for group convolution (denote as GF for short). The depth of the network mainly depends on the N and L, while the width of the network mainly depends on the K and GF. From the existing literature, it is not conducive to the performance of the network, if the number of these variables is too small or too large. Therefore, we validate their effects to a reasonable extent by controlling one variable with other variables fixed on the Pavia Centre data set with the scale factor ×2. As depicted in Fig. 6, we display the curves of the training procedure of different settings in both spatial and spectral reconstruction. Overall, larger N, L, K or GF would lead to better performance in both PSNR and SAM criteria. This is mainly because deeper and wider networks can exploit more hierarchical features and allow more information to pass through. Specifically, we find that the effects of N, L and K on performance are limited. For instance, it can be obviously observed that very narrow layers (i.e., $K = 16$) can converge to almost the same state as larger K. This may be explained by the fact that the hyperspectral training samples are scarce, and the model with large capacity is inevitably prone to overfitting, though it can model more complicated mapping. On the contrary, larger GF can achieve obvious superior results and GF has a greater effect on the network performance, which indicates the importance of low-level features of variety for the reconstruction of HSI. Given the above, we set $N = 4$, $L = 6$, $K = 16$, and GF = 128 in the rest of the experiments.

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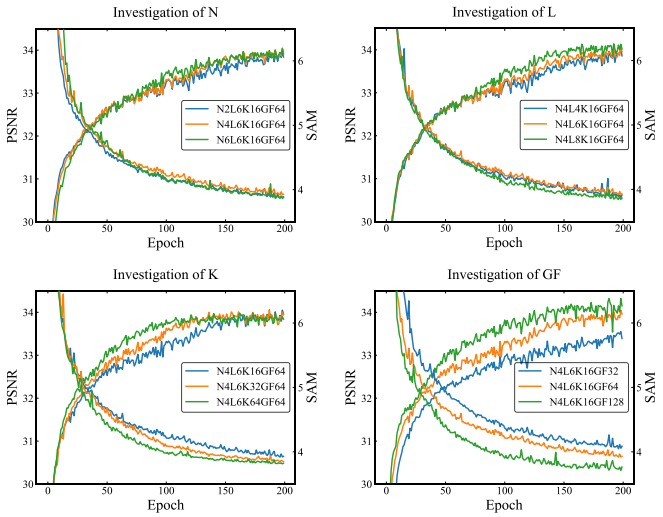


Fig. 6. Parameter analysis of SGARDN with different values of N, L, K and GF. The curves of PSNR criterion are in an upward trend with epochs, while that of SAM criterion are in a downward trend.

3) Ablation Investigation: Table II presents the ablation investigation on the effects of the SGIM and spectral attention module. Bold indicates the best performance. Compared to SGARDN, SGARDN_NSG replaces the SGIM with normal convolution layers that have similar number of parameters. Specifically, SGIM employs three convolutional layers to extract shallow features, of which two are 3×3 group convolutions for intra- and inter-group feature extraction, and the rest is 1×1 convolution. Similarly, SGARDN_NSG employs two 3×3 convolutions and one 1×1 convolution in shallow feature extraction for a fair comparison. Since it treats the input as one group, normal convolution is used instead of group convolution. SGARDN_NSA removes the spectral attention module, and SGARDN_NSGA has neither of these modules. The four networks have the same RDBs. We see that the baseline (SGARDN_NSGA) has been relatively successful, which indicates that taking 3-D HSI as input rather than a stack of 2-D bandwise images enables the network to learn spectral correlations automatically and the effectiveness of the RDB for the spatial-spectral feature exploration. When adopting the SGIM or spectral attention (the middle two rows of Table II), they both achieve gains on all criteria compared to the baseline, especially in the spectral consistency. We can validate that each module can take advantage of spectral correlation to achieve better spatial and spectral reconstruction. When these modules are combined (i.e., SGARDN), the performance is further improved by a large margin. This indicates that these modules are complementary in learning effective spatial and spectral representations simultaneously.

The proposed spectral attention module improves commonly used one by introducing covariance transformation (COV) and depth-wise convolution ($\text{Conv}_{\text{depth}}$). Table III shows the effects of these two improvements. More specifically, the baseline (first row) is obtained by first-order channel attention, using global average pooling on feature maps to generate channel-wise descriptors. We then add the covariance transformation, but use average pooling on covariance statistics to generate channel-wise descriptors (second row). We further add two components to the baseline (third row), that is, use depthwise convolution on covariance statistics to adaptively generate channel-wise descriptors. We can validate that these components can efficiently improve the performance of the baseline. This is mainly because covariance transformation can explore spectral correlations and capture more discriminative representations than first-order pooling, while depthwise convolution can adaptively find more suitable spectral descriptors from covariance statistics than fixed average pooling.

In addition, we demonstrate the superiority of the global hierarchical feature fusion. By removing the skip connections between the N RDSABs, the network is thus impossible to concatenate all levels of features, denoted as SGARDN_NFF. We report results of SGARDN and SGARDN_NFF on the Pavia Centre data set for $2\times$ SR by using $GF = 64$ so that they have roughly the same number of parameters. An improvement is observed in PSNR/SSIM/SAM (33.91/0.9952/3.9295 versus 34.05/0.9954/3.9220). This indicates that the SR reconstruction performance can be improved by the combination of hierarchical features, which has also been demonstrated to be effective in [16] and [17].

4) Loss Function: The MAE loss function and the joint loss function with different balance factors are employed to train our SGARDN, respectively. The quantitative comparisons are shown in Table IV. It is noticed that the performance of joint training varies with the balance factor. The additional spectral loss based on the SAM criterion can further reduce the value of SAM, but at the cost of spatial performance (a decline in the value of PSNR). Due to the existence of a balance

TABLE II

ABLATION INVESTIGATION OF THE PROPOSED MODULES FOR SCALE FACTOR ×2 ON THE PAVIA CENTRE DATA SET

Architecture	PSNR	SSIM	SAM
SGARDN_NSGA	33.90	0.9952	4.0442
SGARDN_NSA	33.98	0.9953	3.8587
SGARDN_NSG	34.01	0.9953	3.7590
SGARDN	**34.32**	**0.9957**	**3.7126**

TABLE III

ABLATION STUDY OF THE PROPOSED SPECTRAL ATTENTION MODULE FOR SCALE FACTOR ×2 ON THE PAVIA CENTRE DATA SET

COV	$Conv_{depth}$	PSNR/SSIM/SAM
×	×	34.21/0.9955/3.7442
√	×	34.26/0.9956/3.7336
√	√	**34.32/0.9957/3.7126**

TABLE IV

LOSS FUNCTION ANALYSIS FOR SCALE FACTOR ×2 ON THE PAVIA CENTRE DATA SET

Loss	PSNR	SSIM	SAM
L_1	**34.32**	**0.9957**	3.7126
$L_1 + \lambda L_{spectral} (\lambda = 0.1)$	34.18	0.9955	3.5532
$L_1 + \lambda L_{spectral} (\lambda = 0.5)$	33.56	0.9948	**3.5245**

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TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE CAVE DATA SET

Scale factor			Bicubic	SRCNN	SRDensenet	RDN	3DFCNN	GDRRN	SSPSR	SGARDN
Cave	2	PSNR	40.71	41.76	42.64	43.57	42.03	42.65	43.15	**44.36**
		SSIM	0.9956	0.9962	0.9965	0.9972	0.9964	0.9969	0.9971	**0.9980**
		SAM	3.7681	3.3306	2.9876	2.9136	3.0350	3.5035	3.0387	**2.8216**
	4	PSNR	33.29	33.32	34.40	36.84	34.02	34.34	36.04	**37.11**
		SSIM	0.9778	0.9796	0.9798	0.9819	0.9805	0.9800	0.9843	**0.9852**
		SAM	5.3821	4.6519	4.4272	3.9462	4.4461	5.1861	4.1514	**3.9050**
	8	PSNR	29.11	29.07	29.21	31.58	29.33	29.85	30.18	**32.00**
		SSIM	0.9449	0.9442	0.9459	0.9555	0.9477	0.9486	0.9521	**0.9559**
		SAM	7.2046	6.7493	6.3802	5.5315	5.8262	6.7658	6.0563	**5.5094**

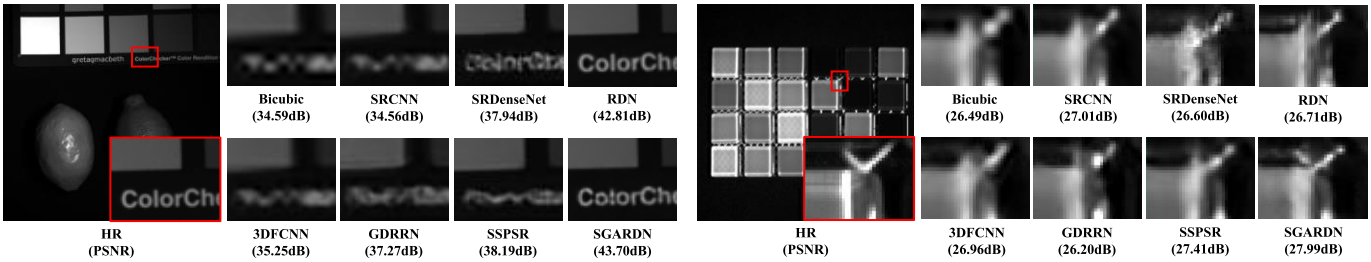


Fig. 7. Visual results of spatial SR on two selected test images of CAVE data set for the 12th band with the scale factor ×4. (Left) Fake and real lemons image. (Right) Glass tiles image.

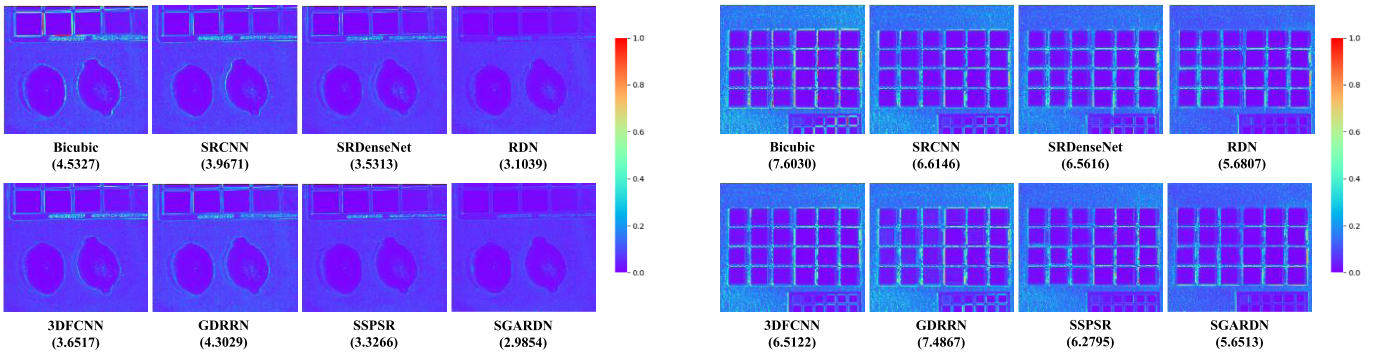


Fig. 8. Visual results of spectral consistency on two selected test images of CAVE data set with the scale factor ×4. (Left) Fake and real lemons image. (Right) Glass tiles image. (The parenthetical content indicates the value of SAM.).

factor that controls the contribution of spatial and spectral constraints, it needs to be constantly adjusted to get better results. Furthermore, training a network becomes nontrivial when the loss function is complex. Therefore, we train the network with only MAE loss in other experiments.

C. Comparisons to State of the Arts

1) Experiments on Synthesized HSIs From CAVE Data Set: Table V tabulates the quantitative results with different scale factors. It is observed that our proposed SGARDN significantly outperforms all the other competing methods with the best PSNR, SSIM, and SAM values for all the cases in the CAVE data set. When the scale factor becomes larger (e.g., ×4 and ×8), our method still holds a similar advantage over the competing ones, whose lead has been shrinking fast because less visual information is available for the SR process. The quantitative comparison of the natural image CNN-based methods demonstrates that the performance is better as the depth increases. This is because deeper networks can significantly improve the quality of representations learned. As for the CNN-based HSI SR methods, 3DFCNN and GDRRN are still difficult to achieve superior results due to the limitation of model capacity. SSPSR employs the spatial-spectral block, consisting of RBs and a spectral attention module, and thus can well exploit the spatial-spectral information (it achieves superior results than 3DFCNN and GDRRN). On the other hand, the HSIs in the CAVE data set have much finer spatial quality and fewer spectral bands than the real-scenario HSIs, more similar to natural images. That is why the CNNs for the SR of natural images still work well. It is worth noting that our method achieves a comparable result with RDN at a much smaller depth, which indicates that the full utilization of HSI characteristics enables the network to have a great potential to promote the process of HSI SR.

The above quantitative analysis is also supported by the qualitative results displayed in Figs. 7–9. In terms of spatial

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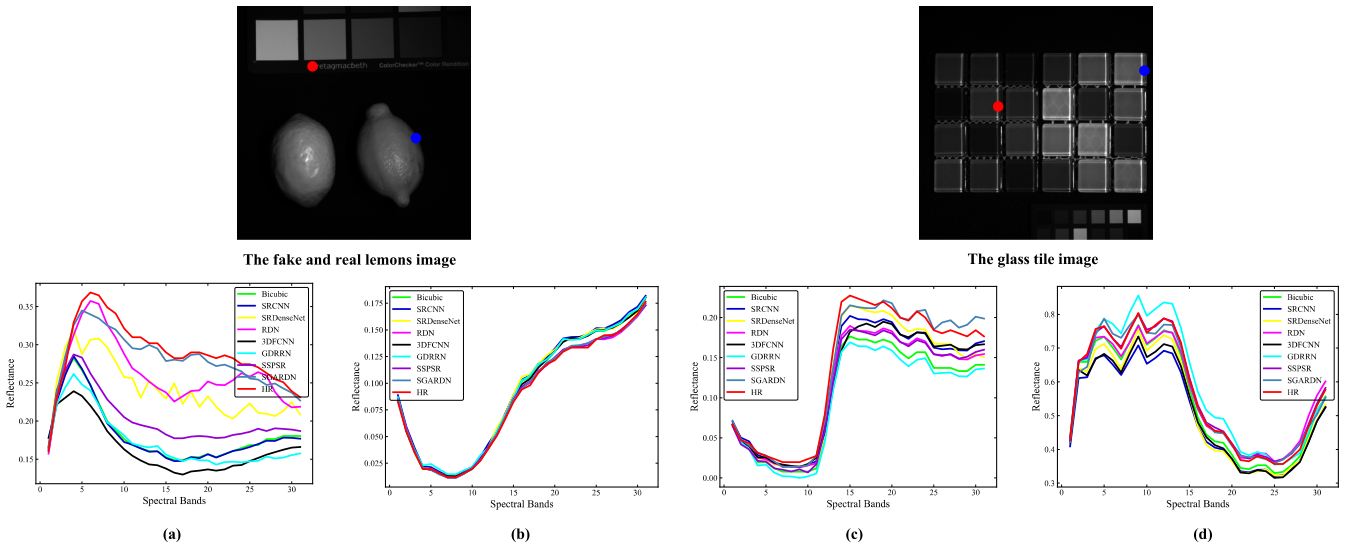


Fig. 9. Example spectral curves on two selected test images of CAVE data set with the scale factor ×4. (Top) Locations. (Bottom) Spectral curves. (a) and (c) Spectrum of red point. (b) and (d) Spectrum of blue point.

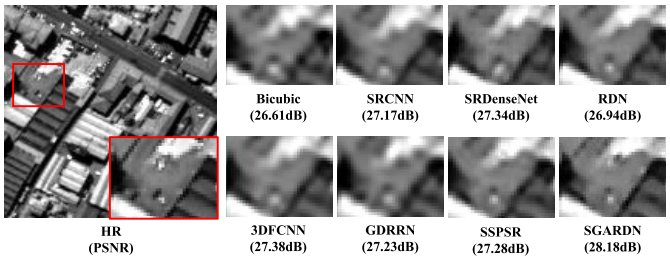


Fig. 10. Visual results of spatial SR on the Pavia Centre data set for the 74th band with the scale factor ×3.

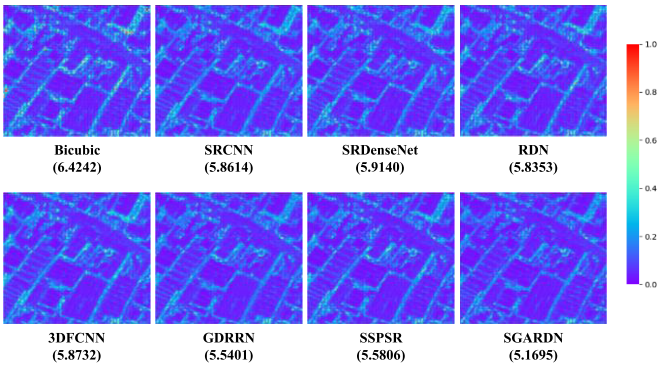


Fig. 11. Visual results of spectral consistency on the Pavia Centre data set with the scale factor ×3. (The parenthetical content indicates the value of SAM.).

reconstruction, the proposed method produces much sharper edges without obvious artifacts than other competing methods. Fig. 7 enlarge the same area marked by the red rectangle to show more details on the super-resolved output of two test images from the CAVE data set, respectively. As can be observed, the proposed method reconstructs more accurate results, such as the details of text in the left part and the connection pattern of glass tiles in the right part, while other methods fail to restore these information. Note that although

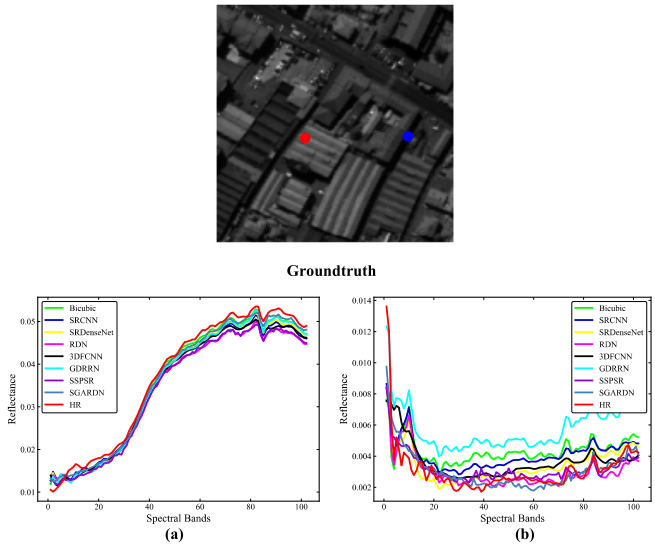


Fig. 12. Example spectral curves on the Pavia Centre data set with the scale factor ×3. (Top) Locations. (Bottom) Spectral curves. (a) Spectrum of red point. (b) Spectrum of blue point.

RDN produces visually appealing output for the fake and real lemons image, it has trouble with the glass tiles image, like other methods. This is likely due to the complex details of the image and the lack of similar texture structures in the training set. For the spectral reconstruction quality, we visualize the SAM to intuitively display the consistency of each spectrum at pixel-levels in Fig. 8. We can find that our proposed method achieves the best spectral fidelity, while other competing methods result in some obvious spectral distortion. Since the SAM still has some limitations [52], we show some spectral curves in Fig. 9 as a supplement. It is demonstrated that our proposed method preserves spectral features better than the other methods, as the spectra reconstructed by the proposed method is closer to the corresponding ground-truth.

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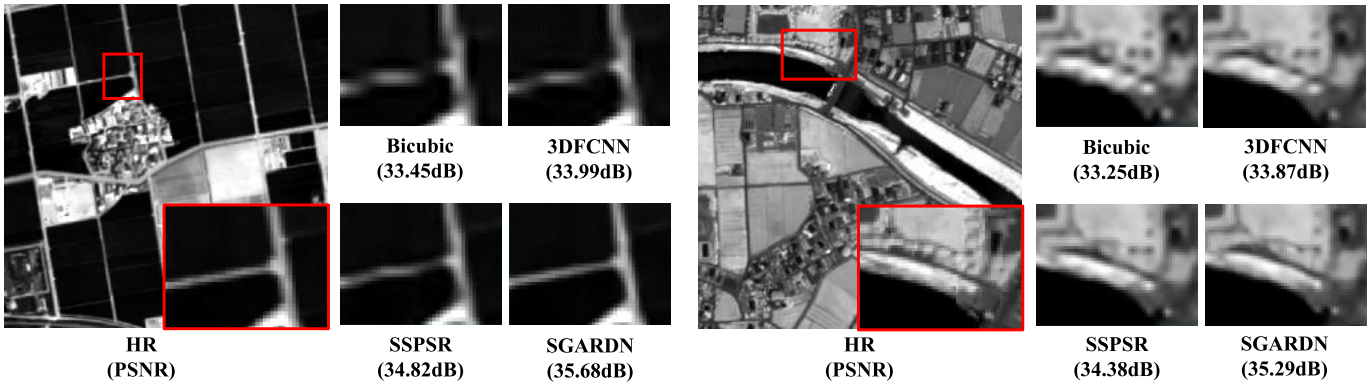


Fig. 13. Visual results of spatial SR on two selected test areas of Chikusei data set with the scale factor ×3. (Left) For the 32th band. (Right) For the 100th band.

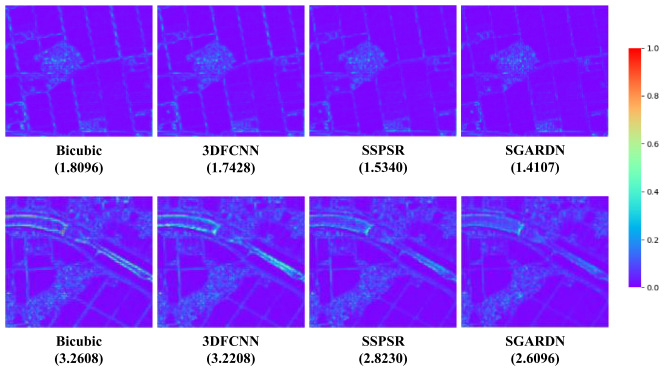


Fig. 14. Visual results of spectral consistency on two selected test areas of Chikusei data set with the scale factor ×3. (The parenthetical content indicates the value of SAM.).

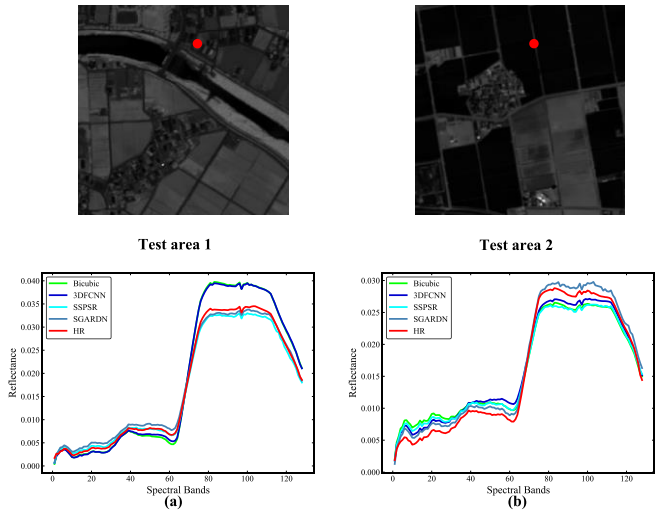


Fig. 15. Example spectral curves on two selected test areas of Chikusei data set with the scale factor ×3. (a) Spectrum of the red point in Test area 1. (b) Spectrum of the red point in Test area 2.

2) Experiments on the Real-Scenario HSI From Pavia Centre Data Set: The quantitative and qualitative results of Pavia Centre are displayed in Table VI and Figs. 10−12, respectively. We have the following observations. As shown in Table VI, our proposed method is superior to other methods in all evaluation criteria with different scale factors.

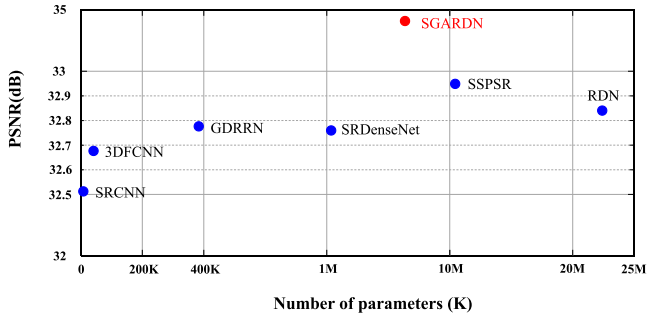


Fig. 16. Number of network parameters versus performance.

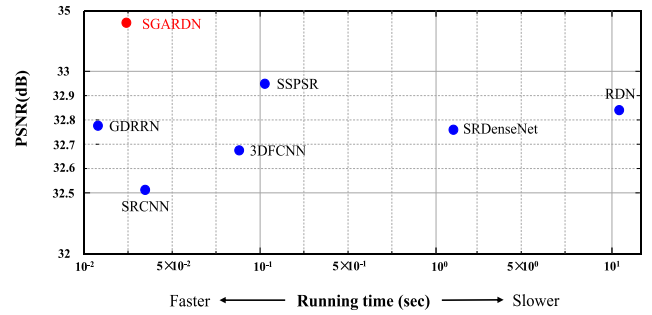


Fig. 17. Running time versus performance.

Compared with the results of CAVE data set, the CNNs for the SR of natural images no longer have advantages over the CNN-based HSI SR methods on the Pavia Centre data set. This presumably due to the following reasons: firstly, they cannot cope with the problem of insufficient detailed information and the very high spectral dimensionality of real-scenario HSIs at the same time; secondly, they are prone to overfitting as a result of large capacity (e.g., RDN). Instead, it is the consideration of spectral correlation (e.g., 3-D convolution, SAM loss, and spatial-spectral block) that leads to better results for the CNN-based HSI SR methods (i.e., 3DFCNN, GDRRN, and SSPSR).

To evaluate the perceptual quality, Fig. 10 shows the super-resolved output of test areas from Pavia Centre and enlarges the same area marked by the red rectangle to show in detail. The proposed method generates more realistic visual results, such as some small objects, whereas other competing

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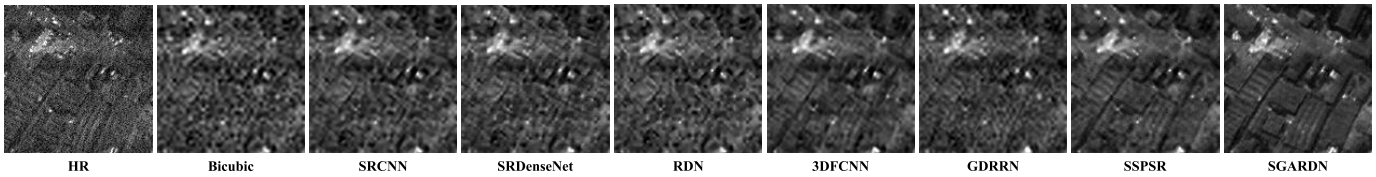


Fig. 18. Visual results of spatial SR on the Pavia Centre data set for the first band with the scale factor $\times 3$.

TABLE VI
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE PAVIA CENTRE DATA SET

Scale factor			Bicubic	SRCNN	SRDensenet	RDN	3DFCNN	GDRRN	SSPSR	SGARDN
Pavia Centre	2	PSNR	31.85	32.51	32.76	32.83	32.68	32.78	32.95	**34.39**
		SSIM	0.9923	0.9934	0.9938	0.9939	0.9937	0.9937	0.9940	**0.9957**
		SAM	4.8810	4.2317	4.1287	4.0248	4.2557	3.8603	4.0701	**3.7044**
	3	PSNR	27.27	27.83	27.96	27.64	28.04	27.86	28.00	**28.71**
		SSIM	0.9775	0.9801	0.9807	0.9792	0.9811	0.9802	0.9808	**0.9839**
		SAM	6.4242	5.8614	5.9140	5.8353	5.8732	5.5401	5.5806	**5.1695**
	4	PSNR	25.52	26.14	26.17	26.03	26.27	26.26	26.19	**26.48**
		SSIM	0.9659	0.9701	0.9702	0.9690	0.9709	0.9707	0.9702	**0.9726**
		SAM	7.5747	6.6941	6.9283	6.8750	6.6870	6.3032	6.4173	**6.1484**

methods fail to recover these details. Regarding the spectrum, Figs. 11 and 12 exhibit the visualization of the SAM criterion and the spectral profile at selected pixels, respectively. Some spectral distortions are found in reconstructed results of existing methods, whereas the spectral curves generated by our method are more consistent with the reference.

3) Experiments on the Real-Scenario HSI From Chikusei Data Set: We further conduct experiments on the Chikusei data set to verify the performance of the proposed method on real-scenario HSIs from different sensors. As stated above, the CNNs for the SR of natural images are relatively inferior in HSI SR. We therefore simply choose two stable methods (i.e., Bicubic and 3DFCNN) and the latest advanced method (i.e., SSPSR) for comparison. From the quantitative results in Table VII, the performance of our proposed method is obviously better than the other competing methods in both spatial and spectral measurements. It can be seen that the average PSNR of our method is 1.21 and 0.87 dB larger than that of Bicubic, while the average SAM is 0.40 and 0.50 smaller than that of Bicubic with $3\times$ and $4\times$ scale factors, respectively, which indicates our model still performs well for larger scale factors in both spatial and spectral reconstruction.

Fig. 13 shows the visual exhibition of the three times reconstructed HSIs. The results of Bicubic interpolation and 3DFCNN are similar, both of which are blurry. SSPSR can recover some details, but still produce blurred edges. In contrast, ours have clearer feature boundaries and restore almost the full details of roads. In Figs. 14 and 15, we visualize the SAM criterion and display some spectral curves in different areas to compare the spectral similarity between HR and SR images in an intuitive way. We can see that the proposed method obtains lower spectral error results and spectral curves closer to the corresponding ground-truth, which indicates the spectral information is well preserved by our method.

4) Model Parameters and Running Time: We show the reconstruction performance (in terms of PSNR) versus the number of network parameters and the running time (the average of 10 evaluations) on the Pavia Centre data set for $2\times$ SR in Figs. 16 and 17, respectively. The proposed SGARDN achieves the highest quality, whilst maintains the competitive number of parameters, in comparison to the state-of-the-art methods (i.e., SSPSR and RDN). SSPSR adopts a wide residual network with 256 filters per convolutional layer. RDN has 16 RDBs, eight convolutional layers per RDB and 64 filters per convolutional layer, leading to a very deep network with a high number of parameters. Therefore, they are hard to stabilize the training procedure and prone to overfitting as a result of large capacity. On the contrary, methods (e.g., 3DFCNN and GDRRN) are difficult to achieve superior results due to the limitation of model capacity.

TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE CHIKUSEI DATA SET

Scale factor			Bicubic	3DFCNN	SSPSR	SGARDN
Chikusei	2	PSNR	36.59	37.07	37.63	**38.16**
		SSIM	0.9971	0.9973	0.9976	**0.9979**
		SAM	1.9079	1.8042	1.7327	**1.6439**
	3	PSNR	31.25	31.62	32.00	**32.46**
		SSIM	0.9899	0.9905	0.9911	**0.9919**
		SAM	3.0329	3.0147	2.7926	**2.6323**
	4	PSNR	28.88	29.32	29.67	**29.75**
		SSIM	0.9830	0.9842	0.9852	**0.9854**
		SAM	4.2139	3.9668	3.8513	**3.7173**

We profile the running time of all the algorithms on the same machine with 2.20 GHz Intel Xeon Silver 4210 CPU (64-GB RAM) and Nvidia GeForce RTX 2080Ti GPU. As shown in Fig. 17, the speed of our SGARDN is faster than all the competing methods except GDRRN. Since the CNNs of natural images SR are applied in a bandwise manner, we set the batch size as the number of spectral bands to realize SR of all bands simultaneously. As a result, they have no advantage in the speed of processing HSIs. It is worth mentioning that the speed of 3DFCNN is mainly limited by the following two reasons. First, 3DFCNN extracts features from the interpolated LR images (i.e., the SR operation is performed in HR space), which increases computation complexity

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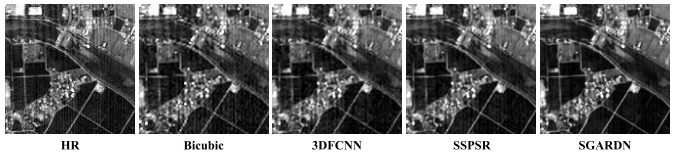


Fig. 19. Visual results of spatial SR on the Chikusei data set for the third band with the scale factor $\times 3$.

quadratically. Second, the computational complexity of 3-D convolution is very large. For instance, both the filter and the output of 3-D convolution have one more dimension than that of 2-D convolution (assuming they are K and T, respectively). Therefore, the number of operations of 3-D convolution is $K \times T$ times that of 2-D convolution.

D. Further Discussion

In general, certain bands of real-scenario HSIs often suffer from severe degradation problems due to sensor malfunction and atmospheric interference, etc. As shown in Figs. 18 and 19, the first band of Pavia Centre and the third band of Chikusei are seriously polluted by different kinds of noise, difficult to distinguish various ground objects. The Bicubic interpolation reconstruction result is blurry and oversmooth. Although other competing methods can increase the spatial resolution to a certain extent, they still have the same problems as the Bicubic interpolation. However, SGARDN can restore details and remove noises of the entire image, producing a visually appealing output. We speculate that this is because our method can exploit complementary information from adjacent bands to deal with severely degraded bands and surpass features that are not conducive to reconstruction through the spectral attention mechanism. Meanwhile, SGARDN can well preserve spectral information according to the aforementioned analysis in Section IV-C. Therefore, we believe that our method is promising in joint denoising and SR of HSI.

V. Conclusion

In this article, we present a novel CNN-based HSI SR approach. The proposed architecture incorporates the spectral grouping strategy and spectral attention mechanism into residual dense network design. Such a grouping strategy, associated with the spectral correlation of HSI, facilitates the capacity of modeling all bands. In this way, the informative spatial features are effectively extracted by group convolutions, while the spectral disorder is alleviated. We further integrate holistic spectral information at earlier stages guided by the spectral attention for more representative feature learning. The residual dense network design is for characterizing deep spatial-spectral features and combining hierarchical features. When equipped with the spectral attention derived from covariance statistics, it enables spectral correlation and discriminative feature learning to boost reconstruction performance. Sufficient ablation investigations verify the effectiveness of each module in spatial resolution enhancement and spectral consistency. The further experimental results of proposed SGARDN on both synthesized and real-scenario HSIs outperforms the competing methods in terms of evaluation criteria and visual effects.

In the future work, we will investigate more efficient learning structures (e.g., semisupervised and unsupervised learning) to deal with the limited number of training samples and the differences of HSIs obtained by different sensors.

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